Derivation of pressurant blowdown thermodynamics

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1 State definitions

The system contains two fluid domains connected by a pressure regulator:

1. Source domain $\mathcal{B}$: the high-pressure storage bottle, referred to throughout as the "bottle".
2. Receiver domain $\mathcal{U}$: the oxidizer tank ullage cavity, referred to as "ullage".

Each domain contains the same pressurant gas, and is described by pressure, temperature, volume, and density, (P, T, V, ρ), with a variable of loaded mass $m = \rho V$. The domains evolve from an initial state $t = 0$ to a final state $t = t_{\text{burn}}$.

We wish to calculate the following:

- m_{press}: initial mass of the pressurant in its tank,
- m_{burn}: the mass of the pressurant gas expelled from the pressurant vessel into the oxidizer tank ullage cavity during the burn,
- m_{pad}: the mass of pressurant gas in the oxidizer tank ullage dome before ignition, and
- V_{bottle}: the required pressurant bottle volume.

Let the thermodynamic variables be defined as follows:

– Initial quantities are marked with $(\cdot)_0$, final quantities with $(\cdot)_\mathrm{f}$

– P_0: Initial bottle storage pressure (200 bar)

– P_f: Final bottle cutoff pressure (90 bar)

– ΔP_margin: Regulator operating margin (20 bar)

– P_ox: Regulated oxidizer tank pressure (70 bar)

– P_sat: Oxidizer saturation vapor pressure (57.5 bar)

– T_0: Ambient initial temperature (293.15 K)

– $V_\mathrm{ox,out}$: Liquid oxidizer volume expelled during burn

– V_{u0}: Initial ullage volume

– u: Initial ullage volume fraction (V_{u0}/V_ox)

– Z: Real-gas compressibility factor (1.15 for N_2 at 200 bar)

– γ: Specific heat ratio of pressurant gas ($\gamma = 1.40$)

– R: Specific gas constant of pressurant (296.8 $\mathrm{J/(kg \cdot K)}$)

– f_reserve: Pressurant reserve margin factor (0.10, i.e., 10%)

2 Closed-form isentropic blowdown of the storage bottle

2.1 Bottle state evolution

Assume rapid blowdown without external wall heat transfer. Gas inside the bottle expands reversibly and adiabatically. The state follows the isentropic path:

$$PV^\gamma = \text{const.}, \quad \rho = \frac{m}{V}$$

$$P(t) \cdot \rho(t)^{-\gamma} = \text{const.}$$

The instantaneous density $\rho(P)$ and temperature $T(P)$ obey:

$$\rho(P) = \rho_0 \left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}}$$
$$T(P) = T_0 \left(\frac{P}{P_0}\right)^{\frac{\gamma-1}{\gamma}}$$

2.2 Delivered mass fraction

Let V_bottle be the internal volume of the storage bottle. The initial gas mass m_0 and final residual gas mass m_f are:

$$m_0 = \rho_0 V_\mathrm{bottle}$$

$$m_\mathrm{f} = \rho(P_\mathrm{f}) V_\mathrm{bottle}$$

Define the delivered gas fraction $f_{\text{delivered}}$:

$$f_{\text{delivered}} = \frac{m_0 - m_{\text{f}}}{m_0} = 1 - \frac{\rho(P_{\text{f}})}{\rho_0}$$

Substitute the isentropic density ratio:

$$f_{\text{delivered}} = 1 - \left(\frac{P_{\text{f}}}{P_0}\right)^{\frac{1}{\gamma}}$$

For $P_0 = 200\,\text{bar}$, $P_{\text{f}} = 90\,\text{bar}$, $\gamma = 1.40$:

$$\begin{aligned} f_{\text{delivered}} &= 1 - \left(\frac{90}{200}\right)^{\frac{1}{1.40}} \\ &= 1 - 0.5652 = 0.4348 \quad (43.48\%) \end{aligned}$$

2.3 Mean delivered inflow temperature

The pressure regulator throttles the gas isenthalpically. Nitrogen behaves as an ideal gas across this pressure range. Isenthalpic throttling of an ideal gas preserves temperature:

$$h_{\text{in}}(t) = h_{\text{bottle}}(t) \implies T_{\text{in}}(t) = T(P(t))$$

The mass-averaged temperature $\bar{T}_{\text{in}}$ of all delivered gas is:

$$\bar{T}_{\text{in}} = \frac{1}{m_0 - m_{\text{f}}} \int_{m_{\text{f}}}^{m_0} T(m)\,\mathrm{d}m$$

Express differential mass $\mathrm{d}m$ in terms of bottle pressure P:

$$\begin{aligned} m(P) &= m_0 \left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}} \\ \mathrm{d}m &= \frac{m_0}{\gamma P_0} \left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}-1} \mathrm{d}P \end{aligned}$$

Substitute $T(P)$ and $\mathrm{d}m$ into the product:

$$\begin{aligned} T(P)\,\mathrm{d}m &= \left[T_0 \left(\frac{P}{P_0}\right)^{\frac{\gamma-1}{\gamma}}\right] \left[\frac{m_0}{\gamma P_0} \left(\frac{P}{P_0}\right)^{\frac{1}{\gamma}-1} \mathrm{d}P\right] \\ &= \frac{m_0 T_0}{\gamma P_0} \left(\frac{P}{P_0}\right)^{\frac{\gamma-1+1-\gamma}{\gamma}} \mathrm{d}P \\ &= \frac{m_0 T_0}{\gamma P_0} \left(\frac{P}{P_0}\right)^{0} \mathrm{d}P = \frac{m_0 T_0}{\gamma P_0}\,\mathrm{d}P \end{aligned}$$

The pressure exponent reduces identically to zero. Evaluate the definite integral from P_f to P_0:

$$\begin{aligned}\int_{m_\mathrm{f}}^{m_0} T\,\mathrm{d}m &= \frac{m_0 T_0}{\gamma P_0}\int_{P_\mathrm{f}}^{P_0}\mathrm{d}P \\ &= \frac{m_0 T_0}{\gamma}\left(1-\frac{P_\mathrm{f}}{P_0}\right)\end{aligned}$$

Divide by total delivered mass:

$$m_0 - m_\mathrm{f} = m_0 f_\mathrm{delivered}$$

This division produces the closed-form mean delivered temperature:

$$\bar{T}_\mathrm{in} = \frac{T_0}{\gamma}\,\frac{1-\left(\frac{P_\mathrm{f}}{P_0}\right)}{1-\left(\frac{P_\mathrm{f}}{P_0}\right)^{\frac{1}{\gamma}}}$$

For $T_0 = 293.15\,\mathrm{K}$, $P_0 = 200\,\mathrm{bar}$, and $P_\mathrm{f} = 90\,\mathrm{bar}$:

$$\begin{aligned}\bar{T}_\mathrm{in} &= \frac{293.15}{1.40}\cdot\frac{1-0.4500}{1-0.5652} \\ &= 209.39\cdot\frac{0.5500}{0.4348} = 264.88\ \mathrm{K}\end{aligned}$$

3 Control volume energy balance on the ullage cavity

3.1 First Law formulation

Define an open control volume encompassing the oxidizer tank ullage space. The boundary expands from state 1 (initial pad gas) to state 2 (end of burn):

- State 1: Pressure P_ox, volume $V_1 = V_{u0}$, and temperature $T_1 = T_0$.
- State 2: Pressure P_ox, volume $V_2 = V_{u0} + V_\mathrm{ox,out}$, temperature T_2.

The First Law of Thermodynamics for this control volume states:

$$U_2 - U_1 = H_\mathrm{in} - W_\mathrm{boundary} + Q$$

1. Internal energy of ideal gas:

$$U_1 = m_1 c_v T_1, \quad U_2 = m_2 c_v T_2$$

2. Enthalpy delivered across the inlet boundary:

$$H_\mathrm{in} = (m_2 - m_1) c_p \bar{T}_\mathrm{in} = m_\mathrm{burn} c_p \bar{T}_\mathrm{in}$$

3. Boundary work performed on the retreating liquid surface:

$$W_{\text{boundary}} = \int_{V_1}^{V_2} P_{\text{ox}}\,\mathrm{d}V = P_{\text{ox}}(V_2 - V_1) = P_{\text{ox}}V_{\text{ox,out}}$$

4. Heat transfer across tank walls:

$$Q \approx 0 \quad \text{(adiabatic during short burn interval)}$$

Substitute these terms into the energy balance:

$$m_2 c_v T_2 - m_1 c_v T_1 = m_{\text{burn}} c_p \bar{T}_{\text{in}} - P_{\text{ox}}V_{\text{ox,out}}$$

3.2 Elimination of internal energies

Apply the ideal gas equation of state $PV = mRT$:

$$m_1 T_1 = \frac{P_{\text{ox}}V_1}{R}, \quad m_2 T_2 = \frac{P_{\text{ox}}V_2}{R}$$

Express specific heat at constant volume in terms of gas constants:

$$c_v = \frac{R}{\gamma - 1}$$

Evaluate internal energies:

$$m_1 c_v T_1 = \left(\frac{P_{\text{ox}}V_1}{R}\right)\left(\frac{R}{\gamma - 1}\right) = \frac{P_{\text{ox}}V_1}{\gamma - 1}$$
$$m_2 c_v T_2 = \left(\frac{P_{\text{ox}}V_2}{R}\right)\left(\frac{R}{\gamma - 1}\right) = \frac{P_{\text{ox}}V_2}{\gamma - 1}$$

The internal energy change ΔU simplifies to:

$$U_2 - U_1 = \frac{P_{\text{ox}}(V_2 - V_1)}{\gamma - 1} = \frac{P_{\text{ox}}V_{\text{ox,out}}}{\gamma - 1}$$

Temperatures T_1 and T_2 vanish completely from ΔU.

3.3 Exact cancellation and invariant pressurant mass

Equate ΔU to the inflow enthalpy and work terms:

$$\frac{P_{\text{ox}}V_{\text{ox,out}}}{\gamma - 1} = m_{\text{burn}} c_p \bar{T}_{\text{in}} - P_{\text{ox}}V_{\text{ox,out}}$$

Collect all terms containing $P_{\text{ox}}V_{\text{liquid}}$ on the left:

$$P_{\text{ox}}V_{\text{ox,out}}\left(1+\frac{1}{\gamma-1}\right)=m_{\text{burn}}c_p\bar{T}_{\text{in}}$$

Simplify the bracketed factor:

$$1+\frac{1}{\gamma-1}=\frac{\gamma}{\gamma-1}$$

Express specific heat at constant pressure:

$$c_p=\frac{\gamma R}{\gamma-1}$$

Substitute c_p into the energy balance:

$$P_{\text{ox}}V_{\text{ox,out}}\left(\frac{\gamma}{\gamma-1}\right)=m_{\text{burn}}\left(\frac{\gamma R}{\gamma-1}\right)\bar{T}_{\text{in}}$$

Divide both sides by the universal factor $\frac{\gamma}{\gamma-1}$:

$$P_{\text{ox}}V_{\text{ox,out}}=m_{\text{burn}}R\bar{T}_{\text{in}}$$

Solve for the burn gas mass m_{burn}:

$$m_{\text{burn}}=\frac{P_{\text{ox}}V_{\text{ox,out}}}{R\bar{T}_{\text{in}}}$$

This equation is exact, the delivered pressurant mass depends strictly on expelled volume, pressure, and mean inflow temperature.

3.4 Scale invariance of final ullage temperature

We compute final ullage temperature T_2 from total mass conservation:

$$m_2=m_1+m_{\text{burn}}$$

Substitute ideal gas mass relations:

$$\frac{P_{\text{ox}}V_2}{RT_2}=\frac{P_{\text{ox}}V_1}{RT_1}+\frac{P_{\text{ox}}V_{\text{ox,out}}}{R\bar{T}_{\text{in}}}$$

Divide the entire equation by P_{ox}/R:

$$\frac{V_2}{T_2}=\frac{V_1}{T_1}+\frac{V_{\text{ox,out}}}{\bar{T}_{\text{in}}}$$

Express volumes V_1 and V_2 in terms of initial ullage fraction u_f:

$$
\begin{aligned}
V_{\text{ox}} &= \frac{V_{\text{ox,out}}}{1-u} \\
V_1 &= uV_{\text{ox}} = \frac{u}{1-u} V_{\text{ox,out}} \\
V_2 &= V_1 + V_{\text{ox,out}} = \frac{1}{1-u} V_{\text{ox,out}}
\end{aligned}
$$

Substitute these expressions into the temperature relation:

$$
\left(\frac{1}{1-u}\right) \frac{V_{\text{ox,out}}}{T_2} = \left(\frac{u}{1-u}\right) \frac{V_{\text{ox,out}}}{T_1} + \frac{V_{\text{ox,out}}}{\bar{T}_{\text{in}}}
$$

Divide through by $V_{\text{ox,out}}$:

$$
\frac{1}{(1-u)T_2} = \frac{u}{(1-u)T_1} + \frac{1}{\bar{T}_{\text{in}}}
$$

Multiply by $(1-u)$:

$$
\frac{1}{T_2} = \frac{u}{T_1} + \frac{1-u}{\bar{T}_{\text{in}}}
$$

The expelled volume $V_{\text{ox,out}}$ cancels completely. The final inverse temperature $1/T_2$ is the convex combination of initial inverse pad gas temperature $1/T_1$ and inverse delivered temperature $1/\bar{T}_{\text{in}}$, weighted strictly by the ullage fraction u. The temperature field is scale-invariant.

4 Sizing synthesis and calculation of remaining variables

4.1 Regulator margin and blowdown termination

The pressure regulator requires a minimum differential pressure to maintain a stable downstream pressure P_{ox}. Let ΔP_{margin} be this required regulator operating margin. The blowdown process terminates when the storage bottle pressure drops to the cutoff pressure P_{f}:

$$
P_{\text{f}} = P_{\text{ox}} + \Delta P_{\text{margin}}
$$

For $P_{\text{ox}} = 70\,\text{bar}$ and $\Delta P_{\text{margin}} = 20\,\text{bar}$, the terminal bottle pressure is:

$$
P_{\text{f}} = 70\,\text{bar} + 20\,\text{bar} = 90\,\text{bar}
$$

When bottle pressure reaches P_{f}, the regulator can no longer deliver the design mass flow rate at pressure P_{ox}. All gas remaining in the bottle below P_{f} is unusable residual mass.

4.2 Real-gas compressibility of high-pressure storage

At initial storage pressures ($P_0 \geq 200\,\text{bar}$), molecular volume and intermolecular forces cause nitrogen to deviate from ideal gas behavior. We introduce the compressibility factor $Z(P, T)$:

$$P = Z\rho RT$$

For gaseous nitrogen at $P_0 = 200\,\text{bar}$ and $T_0 = 293.15\,\text{K}$, the compressibility factor is $Z \approx 1.15$. The initial storage density ρ_0 is:

$$\rho_0 = \frac{P_0}{ZRT_0}$$

For nitrogen ($R = 296.8\,\text{J}/(\text{kg} \cdot \text{K})$):

$$\rho_0 = \frac{200 \times 10^5}{1.15 \cdot 296.8 \cdot 293.15} = 199.8\,\frac{\text{kg}}{\text{m}^3}$$

Ideal gas theory ($Z = 1.0$) predicts $\rho_0 = 229.8\,\text{kg}/\text{m}^3$. Accounting for $Z = 1.15$ prevents a 13% under-prediction of required bottle volume.

4.3 Pre-pressurization pad gas mass

Before motor ignition, the oxidizer tank contains liquid oxidizer and an initial ullage dome:

$$V_1 = V_{u0} = uV_{\text{ox}} = \frac{u}{1-u}V_{\text{ox,out}}$$

Liquid oxidizer evaporates until its vapor reaches saturation pressure P_{sat}. By Dalton's law of partial pressures, the total ullage pressure is:

$$P_{\text{ox}} = P_{\text{sat}} + P_{\text{pad}}$$

The partial pressure that pressurant gas must supply is:

$$P_{\text{pad}} = \max(0,\, P_{\text{ox}} - P_{\text{sat}})$$

At ambient temperature T_0, the required pad gas mass is:

$$m_{\text{pad}} = \frac{P_{\text{pad}} V_{u0}}{RT_0} = \frac{\max(0,\, P_{\text{ox}} - P_{\text{sat}})V_{u0}}{RT_0}$$

If $P_{\text{ox}} \leq P_{\text{sat}}$, oxidizer vapor provides all required pad pressure, and $m_{\text{pad}} = 0$.

4.4 Total loaded pressurant mass

The total pressurant gas mass delivered into the oxidizer tank during pre-launch and flight is:

$$m_{\text{delivered}} = m_{\text{pad}} + m_{\text{burn}}$$

From Section 2.2, the bottle delivers only fraction $f_{\text{delivered}}$ of its initial gas:

$$f_{\text{delivered}} = 1 - \left(\frac{P_{\text{f}}}{P_0}\right)^{\frac{1}{\gamma}}$$

We apply an engineering reserve factor $f_{\text{reserve}} = 0.10$ (10% margin). The total initial pressurant mass loaded into the bottle on the launch pad is:

$$\begin{aligned} m_{\text{press}} &= (1 + f_{\text{reserve}}) \frac{m_{\text{delivered}}}{f_{\text{delivered}}} \\ &= 1.10 \cdot \frac{m_{\text{pad}} + m_{\text{burn}}}{1 - (P_{\text{f}}/P_0)^{\frac{1}{\gamma}}} \end{aligned}$$

4.5 Storage bottle internal volume

With total loaded mass m_{press} and real initial density ρ_0, the required internal volume of the storage bottle is:

$$V_{\text{bottle}} = \frac{m_{\text{press}}}{\rho_0} = \frac{m_{\text{press}} \cdot ZRT_0}{P_0}$$

This volume feeds directly into the layered pressure vessel sizing algorithm in `modules/BEARS_Rocket/comp_tank.py`.

4.6 Residual pressurant mass at burnout

The unusable gas trapped inside the bottle at cutoff is:

$$\begin{aligned} m_{\text{residual}} &= m_{\text{press}} - m_{\text{delivered}} \\ &= m_{\text{press}} \left(1 - \frac{f_{\text{delivered}}}{1 + f_{\text{reserve}}}\right) \end{aligned}$$

This trapped dead gas mass is carried to apogee. It is added to the vehicle dry mass budget in `modules/BEARS_Rocket/comp_mass.py`

5 Summary of computational pipeline

The complete computational pipeline executes as follows:

1. **Regulator Cutoff Pressure:**

$$P_{\mathrm{f}} = P_{\mathrm{ox}} + \Delta P_{\mathrm{margin}}$$

2. **Delivered Fraction & Inflow Temperature:**

$$f_{\mathrm{delivered}} = 1 - \left(\frac{P_{\mathrm{f}}}{P_0}\right)^{\frac{1}{\gamma}}$$
$$\bar{T}_{\mathrm{in}} = \frac{T_0}{\gamma}\,\frac{1-(P_{\mathrm{f}}/P_0)}{1-(P_{\mathrm{f}}/P_0)^{\frac{1}{\gamma}}}$$

3. **Ullage Burn Mass:**

$$m_{\mathrm{burn}} = \frac{P_{\mathrm{ox}} V_{\mathrm{ox,out}}}{R\bar{T}_{\mathrm{in}}}$$

4. **Pad Gas Mass:**

$$m_{\mathrm{pad}} = \frac{\max(0,\, P_{\mathrm{ox}} - P_{\mathrm{sat}}) V_{u0}}{RT_0}$$

5. **Total Loaded Pressurant Mass:**

$$m_{\mathrm{press}} = 1.10 \cdot \frac{m_{\mathrm{burn}} + m_{\mathrm{pad}}}{f_{\mathrm{delivered}}}$$

6. **Storage Bottle Volume:**

$$V_{\mathrm{bottle}} = \frac{m_{\mathrm{press}} \cdot ZRT_0}{P_0}$$